

- **Device Gateway** – Your devices connect with each other and AWS cloud securely using Device Gateway. Device gateway uses the publication/subscription model for exchanging messages. It gives IoT devices the provision of broadcasting data to multiple subscribers.
- **Authenticate** – IoT implements authentication and encryption at every point of the connection. It supports AWS' own method of authentication and also certificate based authentication. You can create certificates and deploy these certificates for your devices.
- **Registry** – AWS provides Registry for keeping track of your devices. Each and every device has a unique identifier in Registry. And you can also store attributes and capabilities of your devices in form of metadata.
- **Device Shadows** – AWS IoT allows you to create a virtual version of your devices called shadows. Shadows reflect the last reported state and future desired state of your device. Even if the device is offline, you can still get its last reported state from its shadow.
- **Rule Engine** – You can build an IoT application that will collect, analyse, process data and accordingly perform some actions on devices. All of these can be done with the help of rule engine. It will evaluate data and perform task as defined in your rule.

Services of AWS IoT:

- **AWS IoT core** – AWS IoT core is the cloud platform for Internet of Things. It helps to create a secure and reliable connection between your devices and other cloud based AWS services. IoT core can also collect data coming from your devices and accordingly perform some action.
- **AWS IoT Device Management** – AWS IoT device management helps you to manage your devices including tasks such as onboarding and sending remote updates. Devices can be organised into groups and monitored using device management console.
- **AWS Greengrass** – Using AWS Greengrass, you can run local compute, send data through secure connection between devices, even when they are not connected to the internet.
- **AWS IoT Analytics** – You need not worry about the cost and complexity for building your own IoT analytics platform. AWS provides a service for running analytics known as AWS IoT analytics.
- **Amazon FreeRTOS** – It is a FreeRTOS based operating system for small microcontroller based devices. You can securely connect these devices to AWS IoT core or more powerful edge devices.